

RoFormer: ENHANCED TRANSFORMER WITH ROTARY POSITION EMBEDDING

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November 9, 2023

ABSTRACT

Position encoding recently has shown effective in the transformer architecture. It enables valuable supervision for dependency modeling between elements at different positions of the sequence. In this paper, we first investigate various methods to integrate positional information into the learning process of transformer-based language models. Then, we propose a novel method named Rotary Position Embedding(RoPE) to effectively leverage the positional information. Specifically, the proposed RoPE encodes the absolute position with a rotation matrix and meanwhile incorporates the explicit relative position dependency in self-attention formulation. Notably, RoPE enables valuable properties, including the flexibility of sequence length, decaying inter-token dependency with increasing relative distances, and the capability of equipping the linear self-attention with relative position encoding. Finally, we evaluate the enhanced transformer with rotary position embedding, also called RoFormer, on various long text classification benchmark datasets. Our experiments show that it consistently overcomes its alternatives. Furthermore, we provide a theoretical analysis to explain some experimental results. RoFormer is already integrated into Huggingface: https://huggingface.co/docs/transformers/model_doc/roformer.

Keywords Pre-trained Language Models · Position Information Encoding · Pre-training · Natural Language Processing.

1 Introduction

The sequential order of words is of great value to natural language understanding. Recurrent neural networks (RRNs) based models encode tokens' order by recursively computing a hidden state along the time dimension. Convolution neural networks (CNNs) based models (CNNs) Gehring et al. [2017] were typically considered position-agnostic, but recent work Islam et al. [2020] has shown that the commonly used padding operation can implicitly learn position information. Recently, the pre-trained language models (PLMs), which were built upon the transformer Vaswani et al. [2017], have achieved the state-of-the-art performance of various natural language processing (NLP) tasks, including context representation learning Devlin et al. [2019], machine translation Vaswani et al. [2017], and language modeling Radford et al. [2019], to name a few. Unlike, RRNs and CNNs-based models, PLMs utilize the self-attention mechanism to semantically capture the contextual representation of a given corpus. As a consequence, PLMs achieve a significant improvement in terms of parallelization over RNNs and improve the modeling ability of longer intra-token relations compared to CNNs¹.

¹A stack of multiple CNN layers can also capture longer intra-token relation, here we only consider single layer setting.

RoFormer: 增强型Transformer与旋转位置嵌入

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2023年11月9日

摘要

位置编码最近在Transformer架构中显示出有效性。它为序列中不同位置元素之间的依赖建模提供了有价值的监督。在本文中，我们首先研究了将位置信息集成到基于Transformer的语言模型学习过程中的各种方法。然后，我们提出了一种名为旋转位置嵌入(RoPE)的新方法，以有效地利用位置信息。具体来说，所提出的RoPE使用旋转矩阵对绝对位置进行编码，并同时将显式的相对位置依赖关系纳入自注意力公式中。值得注意的是，RoPE具有有价值的特性，包括序列长度的灵活性、随着相对距离的增加而衰减的token间依赖关系，以及为线性自注意力配备相对位置编码的能力。最后，我们在各种长文本分类基准数据集上评估了具有旋转位置嵌入的增强型Transformer，也称为RoFormer。我们的实验表明，它始终优于其替代方案。此外，我们提供了理论分析来解释一些实验结果。RoFormer已经集成到Huggingface: https://huggingface.co/docs/transformers/model_doc/roformer。

关键词 预训练语言模型 · 位置信息编码 · 预训练 · 自然语言处理。

1 引言

词语的顺序对自然语言理解具有很大价值。基于循环神经网络(RRN)的模型通过沿时间维度递归计算隐藏状态来编码token的顺序。基于卷积神经网络(CNN)的模型(CNN) Gehring等人[2017]通常被认为是位置无关的，但最近的工作Islam等人[2020]表明，常用的填充操作可以隐式地学习位置信息。最近，基于Transformer Vaswani等人[2017]，构建的预训练语言模型(PLMs)在各种自然语言处理(NLP)任务中取得了最先进的性能，包括上下文表示学习Devlin等人[2019]，机器翻译Vaswani等人[2017]，和语言建模Radford等人[2019]，等。与RRNs和基于CNNs的模型不同，PLMs利用自注意力机制来语义地捕获给定语料库的上下文表示。因此，PLMs在并行化方面比RNNs有显著提升，并且与CNNs相比，提高了对更长token内关系的建模能力₁。

₁多个卷积神经网络层的堆叠也可以捕获更长的词内关系，这里我们仅考虑单层设置。

It is noteworthy that the self-attention architecture of the current PLMs has shown to be position-agnostic Yun et al. [2020]. Following this claim, various approaches have been proposed to encode the position information into the learning process. On one side, generated absolute position encoding through a pre-defined function Vaswani et al. [2017] was added to the contextual representations, while a trainable absolute position encoding Gehring et al. [2017], Devlin et al. [2019], Lan et al. [2020], Clark et al. [2020], Radford et al. [2019], Radford and Narasimhan [2018]. On the other side, the previous work Parikh et al. [2016], Shaw et al. [2018], Huang et al. [2018], Dai et al. [2019], Yang et al. [2019], Raffel et al. [2020], Ke et al. [2020], He et al. [2020], Huang et al. [2020] focuses on relative position encoding, which typically encodes the relative position information into the attention mechanism. In addition to these approaches, the authors of Liu et al. [2020] have proposed to model the dependency of position encoding from the perspective of Neural ODE Chen et al. [2018a], and the authors of Wang et al. [2020] have proposed to model the position information in complex space. Despite the effectiveness of these approaches, they commonly add the position information to the context representation and thus render them unsuitable for the linear self-attention architecture.

In this paper, we introduce a novel method, namely Rotary Position Embedding(RoPE), to leverage the positional information into the learning process of PLMS. Specifically, RoPE encodes the absolute position with a rotation matrix and meanwhile incorporates the explicit relative position dependency in self-attention formulation. Note that the proposed RoPE is prioritized over the existing methods through valuable properties, including the sequence length flexibility, decaying inter-token dependency with increasing relative distances, and the capability of equipping the linear self-attention with relative position encoding. Experimental results on various long text classification benchmark datasets show that the enhanced transformer with rotary position embedding, namely RoFormer, can give better performance compared to baseline alternatives and thus demonstrates the efficacy of the proposed RoPE.

In brief, our contributions are three-folds as follows:

- We investigated the existing approaches to the relative position encoding and found that they are mostly built based on the idea of the decomposition of adding position encoding to the context representations. We introduce a novel method, namely Rotary Position Embedding(RoPE), to leverage the positional information into the learning process of PLMS. The key idea is to encode relative position by multiplying the context representations with a rotation matrix with a clear theoretical interpretation.
- We study the properties of RoPE and show that it decays with the relative distance increased, which is desired for natural language encoding. We kindly argue that previous relative position encoding-based approaches are not compatible with linear self-attention.
- We evaluate the proposed RoFormer on various long text benchmark datasets. Our experiments show that it consistently achieves better performance compared to its alternatives. Some experiments with pre-trained language models are available on GitHub: <https://github.com/ZhuiyiTechnology/roformer>.

The remaining of the paper is organized as follows. We establish a formal description of the position encoding problem in self-attention architecture and revisit previous works in Section (2). We then describe the rotary position encoding (RoPE) and study its properties in Section (3). We report experiments in Section (4). Finally, we conclude this paper in Section (5).

2 Background and Related Work

2.1 Preliminary

Let $\mathbb{S}_N = \{w_i\}_{i=1}^N$ be a sequence of N input tokens with w_i being the i^{th} element. The corresponding word embedding of $\mathbb{S}_N$ is denoted as $\mathbb{E}_N = \{\mathbf{x}_i\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$ is the d -dimensional word embedding vector of token w_i without position information. The self-attention first incorporates position information to the word embeddings and transforms them into queries, keys, and value representations.

$$\begin{aligned} \mathbf{q}_m &= f_q(\mathbf{x}_m, m) \\ \mathbf{k}_n &= f_k(\mathbf{x}_n, n) \\ \mathbf{v}_n &= f_v(\mathbf{x}_n, n), \end{aligned} \quad (1)$$

where $\mathbf{q}_m, \mathbf{k}_n$ and $\mathbf{v}_n$ incorporate the m^{th} and n^{th} positions through f_q, f_k and f_v , respectively. The query and key values are then used to compute the attention weights, while the output is computed as the weighted sum over the value

值得注意的是, 当前PLMs的自注意力架构表现出位置无关性Yun等人[2020]。基于这一观点, 已经提出了各种方法将位置信息编码到学习过程中。一方面, 通过预定义函数Vaswani等人[2017]生成的绝对位置编码被添加到上下文表示中, 同时Gehring等人[2017], Devlin等人[2019], Lan等人[2020], Clark等人[2020], Radford等人[2019], Radford和Narasimhan [2018]提出了可训练的绝对位置编码。另一方面, 之前的工作Parikh等人[2016], Shaw等人[2018], Huang等人[2018], Dai等人[2019], Yang等人[2019], Raffel等人[2020], Ke等人[2020], He等人[2020], Huang等人[2020]专注于相对位置编码, 通常将相对位置信息编码到注意力机制中。除此之外, Liu等人[2020]的作者从神经常微分方程Chen等人[2018a], 的角度提出了对位置编码依赖性的建模, Wang等人[2020]的作者提出了在复杂数学空间中对位置信息的建模。尽管这些方法很有效, 但它们通常将位置信息添加到上下文表示中, 因此不适合线性自注意力架构。

在本文中, 我们介绍了一种名为旋转位置嵌入(RoPE)的新方法, 用于将位置信息利用到PLMS的学习过程中。具体来说, RoPE使用旋转矩阵对绝对位置进行编码, 同时在自注意力公式中融入显式的相对位置依赖。请注意, 所提出的RoPE通过宝贵的特性优先于现有方法, 包括序列长度灵活性、随着相对距离增加而衰减的token间依赖, 以及为线性自注意力配备相对位置编码的能力。在各种长文本分类基准数据集上的实验结果表明, 带有旋转位置嵌入的增强型Transformer, 即RoFormer, 与基线替代方案相比能够提供更好的性能, 从而证明了所提出的RoPE的有效性。

简而言之, 我们的贡献有三点如下:

- 我们研究了现有的相对位置编码方法, 发现它们大多基于将位置编码添加到上下文表示的分解思想。我们介绍了一种名为旋转位置嵌入(RoPE)的新方法, 用于将位置信息利用到PLMS的学习过程中。关键思想是通过将上下文表示与一个具有明确理论解释的旋转矩阵相乘来编码相对位置。
- 我们研究了RoPE的性质, 并表明它与相对距离的增加成衰减关系, 这对于自然语言编码是理想的。我们恳切地认为, 以前基于相对位置编码的方法与线性自注意力不兼容。
- 我们在各种长文本基准数据集上评估了提出的RoFormer。我们的实验表明, 它始终比其替代方案表现更好。一些使用预训练语言模型的实验可以在GitHub上找到: <https://github.com/ZhuiyiTechnology/roformer>。

本文的其余部分组织如下。我们在第(2)节中建立了自注意力架构中位置编码问题的形式描述, 并回顾了以前的工作。然后在第(3)节中描述了旋转位置编码(RoPE), 并研究了它的性质。我们在第(4)节中报告了实验。最后, 我们在第(5)节中总结了本文。

2 背景和相关工作

2.1 预备知识

Let $\mathbb{S}_N = \{w_i\}_{i=1}^N$ be a sequence of N input tokens with w_i being the i^{th} element. $\mathbb{S}_N$ 的对应词嵌入表示为 $\mathbb{E}_N = \{\mathbf{x}_i\}_{i=1}^N$, 其中 $\mathbf{x}_i \in \mathbb{R}^d$ 是token w_i 的 d 维词嵌入向量, 不包含位置信息。自注意力首先将位置信息结合到词嵌入中, 并将它们转换为查询、键和值表示。

$$\begin{aligned} \mathbf{q}_m &= f_q(\mathbf{x}_m, m) \\ \mathbf{k}_n &= f_k(\mathbf{x}_n, n) \\ \mathbf{v}_n &= f_v(\mathbf{x}_n, n), \end{aligned} \quad (1)$$

在 $\mathbf{q}_m, \mathbf{k}_n$ 和 $\mathbf{v}_n$ 中分别通过 m^{th} 和 n^{th} 位置来整合 f_q, f_k 和 f_v 。然后使用查询和键值来计算注意力权重, 而输出是通过将值进行加权求和来计算的

representation.

$$a_{m,n} = \frac{\exp(\frac{\mathbf{q}_m^\top \mathbf{k}_n}{\sqrt{d}})}{\sum_{j=1}^N \exp(\frac{\mathbf{q}_m^\top \mathbf{k}_j}{\sqrt{d}})} \quad (2)$$

$$\mathbf{o}_m = \sum_{n=1}^N a_{m,n} \mathbf{v}_n$$

The existing approaches of transformer-based position encoding mainly focus on choosing a suitable function to form Equation (1).

2.2 Absolute position embedding

A typical choice of Equation (1) is

$$f_{t:t \in \{q,k,v\}}(\mathbf{x}_i, i) := \mathbf{W}_{t:t \in \{q,k,v\}}(\mathbf{x}_i + \mathbf{p}_i), \quad (3)$$

where $\mathbf{p}_i \in \mathbb{R}^d$ is a d-dimensional vector depending of the position of token $\mathbf{x}_i$. Previous work Devlin et al. [2019], Lan et al. [2020], Clark et al. [2020], Radford et al. [2019], Radford and Narasimhan [2018] introduced the use of a set of trainable vectors $\mathbf{p}_i \in \{\mathbf{p}_t\}_{t=1}^L$, where L is the maximum sequence length. The authors of Vaswani et al. [2017] have proposed to generate $\mathbf{p}_i$ using the sinusoidal function.

$$\begin{cases} \mathbf{p}_{i,2t} &= \sin(k/10000^{2t/d}) \\ \mathbf{p}_{i,2t+1} &= \cos(k/10000^{2t/d}) \end{cases} \quad (4)$$

in which $\mathbf{p}_{i,2t}$ is the $2t^{th}$ element of the d-dimensional vector $\mathbf{p}_i$. In the next section, we show that our proposed RoPE is related to this intuition from the sinusoidal function perspective. However, instead of directly adding the position to the context representation, RoPE proposes to incorporate the relative position information by multiplying with the sinusoidal functions.

2.3 Relative position embedding

The authors of Shaw et al. [2018] applied different settings of Equation (1) as following:

$$\begin{aligned} f_q(\mathbf{x}_m) &:= \mathbf{W}_q \mathbf{x}_m \\ f_k(\mathbf{x}_n, n) &:= \mathbf{W}_k(\mathbf{x}_n + \tilde{\mathbf{p}}_r^k) \\ f_v(\mathbf{x}_n, n) &:= \mathbf{W}_v(\mathbf{x}_n + \tilde{\mathbf{p}}_r^v) \end{aligned} \quad (5)$$

where $\tilde{\mathbf{p}}_r^k, \tilde{\mathbf{p}}_r^v \in \mathbb{R}^d$ are trainable relative position embeddings. Note that $r = \text{clip}(m - n, r_{\min}, r_{\max})$ represents the relative distance between position m and n . They clipped the relative distance with the hypothesis that precise relative position information is not useful beyond a certain distance. Keeping the form of Equation (3), the authors Dai et al. [2019] have proposed to decompose $\mathbf{q}_m^\top \mathbf{k}_n$ of Equation (2) as

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{p}_n + \mathbf{p}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{p}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{p}_n, \quad (6)$$

the key idea is to replace the absolute position embedding $\mathbf{p}_n$ with its sinusoid-encoded relative counterpart $\tilde{\mathbf{p}}_{m-n}$, while the absolute position $\mathbf{p}_m$ in the third and fourth term with two trainable vectors $\mathbf{u}$ and $\mathbf{v}$ independent of the query positions. Further, $\mathbf{W}_k$ is distinguished for the content-based and location-based key vectors $\mathbf{x}_n$ and $\mathbf{p}_n$, denoted as $\mathbf{W}_k$ and $\tilde{\mathbf{W}}_k$, resulting in:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \tilde{\mathbf{W}}_k \tilde{\mathbf{p}}_{m-n} + \mathbf{u}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{v}^\top \mathbf{W}_q^\top \tilde{\mathbf{W}}_k \tilde{\mathbf{p}}_{m-n} \quad (7)$$

It is noteworthy that the position information in the value term is removed by setting $f_v(\mathbf{x}_j) := \mathbf{W}_v \mathbf{x}_j$. Later work Raffel et al. [2020], He et al. [2020], Ke et al. [2020], Huang et al. [2020] followed these settings by only encoding the relative position information into the attention weights. However, the authors of Raffel et al. [2020] reformed Equation (6) as:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + b_{i,j} \quad (8)$$

where $b_{i,j}$ is a trainable bias. The authors of Ke et al. [2020] investigated the middle two terms of Equation (6) and found little correlations between absolute positions and words. The authors of Raffel et al. [2020] proposed to model a pair of words or positions using different projection matrices.

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{p}_m^\top \mathbf{U}_q^\top \mathbf{U}_k \mathbf{p}_n + b_{i,j} \quad (9)$$

表示。

$$a_{m,n} = \frac{\exp(\frac{\mathbf{q}_m^\top \mathbf{k}_n}{\sqrt{d}})}{\sum_{j=1}^N \exp(\frac{\mathbf{q}_m^\top \mathbf{k}_j}{\sqrt{d}})} \quad (2)$$

$$\mathbf{o}_m = \sum_{n=1}^N a_{m,n} \mathbf{v}_n$$

现有的基于Transformer的位置编码方法主要集中于选择一个合适的函数来形成公式(1)。

2.2 绝对位置嵌入

公式(1)的一个典型选择是

$$f_{t:t \in \{q,k,v\}}(\mathbf{x}_i, i) := \mathbf{W}_{t:t \in \{q,k,v\}}(\mathbf{x}_i + \mathbf{p}_i), \quad (3)$$

其中 $\mathbf{p}_i \in \mathbb{R}^d$ 是一个依赖于标记 $\mathbf{x}_i$ 位置的 d 维向量。先前的工作 Devlin等人. [2019],Lan等人 [2020], Clark等人 [2020], Radford等人 [2019], Radford和Narasimhan [2018] 引入了使用一组可训练向量 $\mathbf{p}_i \in \{\mathbf{p}_t\}_{t=1}^L$ 的方法, 其中 L 是最大序列长度。Vaswani等人 [2017] 的作者提出了使用正弦函数生成 $\mathbf{p}_i$ 。

$$\begin{cases} \mathbf{p}_{i,2t} &= \sin(k/10000^{2t/d}) \\ \mathbf{p}_{i,2t+1} &= \cos(k/10000^{2t/d}) \end{cases} \quad (4)$$

其中 $\mathbf{p}_{i,2t}$ 是 d 维向量 $\mathbf{p}_i$ 的 $2t^{th}$ 元素。在下一节中, 我们表明我们提出的RoPE与正弦函数的直觉相关。然而, RoPE并不是直接将位置添加到上下文表示中, 而是通过乘以正弦函数来整合相对位置信息。

2.3 相对位置嵌入

Shaw等人[2019] [2018] 应用了方程(1)的不同设置, 如下:

$$\begin{aligned} f_q(\mathbf{x}_m) &:= \mathbf{W}_q \mathbf{x}_m \\ f_k(\mathbf{x}_n, n) &:= \mathbf{W}_k(\mathbf{x}_n + \tilde{\mathbf{p}}_r^k) \\ f_v(\mathbf{x}_n, n) &:= \mathbf{W}_v(\mathbf{x}_n + \tilde{\mathbf{p}}_r^v) \end{aligned} \quad (5)$$

其中 $\tilde{\mathbf{p}}_r^k, \tilde{\mathbf{p}}_r^v \in \mathbb{R}^d$ 是可训练的相对位置嵌入。请注意, $r = \text{clip}(m - n, r_{\min}, r_{\max})$ 表示位置 m 和 n 之间的相对距离。他们通过假设精确的相对位置信息在某个距离之外没有用, 对相对距离进行了裁剪。保持方程(3)的形式, Dai等人[2019][2019] 提出将方程(2)中的 $\mathbf{q}_m^\top \mathbf{k}_n$ 分解为

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{p}_n + \mathbf{p}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{p}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{p}_n, \quad (6)$$

关键思想是用绝对位置嵌入 $\mathbf{p}_n$ 的正弦编码的相对对应物 $\tilde{\mathbf{p}}_{m-n}$ 来替换, 同时在第三和第四项中的绝对位置 $\mathbf{p}_m$ 用与查询位置无关的可训练向量 $\mathbf{u}$ 和 $\mathbf{v}$ 来替换。此外, $\mathbf{W}_k$ 用于基于内容和基于位置的关键向量 $\mathbf{x}_n$ 和 $\mathbf{p}_n$, 表示为 $\mathbf{W}_k$ 和 $\tilde{\mathbf{W}}_k$, 结果为:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \tilde{\mathbf{W}}_k \tilde{\mathbf{p}}_{m-n} + \mathbf{u}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{v}^\top \mathbf{W}_q^\top \tilde{\mathbf{W}}_k \tilde{\mathbf{p}}_{m-n} \quad (7)$$

值得注意的是, 通过设置 $f_v(\mathbf{x}_j) := \mathbf{W}_v \mathbf{x}_j$ 移除了值项中的位置信息。后续工作 Raffel等人 [2020], He等人 [2020], Ke等人 [2020], Huang等人 [2020] 遵循这些设置, 仅将相对位置信息编码到注意力权重中。然而, Raffel等人 [2020] 的作者重新改写了公式(6)如下:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + b_{i,j} \quad (8)$$

其中 $b_{i,j}$ 是一个可训练偏差。Ke等人 [2020] 研究了公式(6)的中间两项, 发现绝对位置与词之间几乎没有相关性。Raffele等人 [2020] 提出使用不同的投影矩阵来建模一对词或位置。

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{p}_m^\top \mathbf{U}_q^\top \mathbf{U}_k \mathbf{p}_n + b_{i,j} \quad (9)$$

The authors of He et al. [2020] argued that the relative positions of two tokens could only be fully modeled using the middle two terms of Equation (6). As a consequence, the absolute position embeddings $\mathbf{p}_m$ and $\mathbf{p}_n$ were simply replaced with the relative position embeddings $\tilde{\mathbf{p}}_{m-n}$:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \tilde{\mathbf{p}}_{m-n} + \tilde{\mathbf{p}}_{m-n}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n \quad (10)$$

A comparison of the four variants of the relative position embeddings Radford and Narasimhan [2018] has shown that the variant similar to Equation (10) is the most efficient among the other three. Generally speaking, all these approaches attempt to modify Equation (6) based on the decomposition of Equation (3) under the self-attention settings in Equation (2), which was originally proposed in Vaswani et al. [2017]. They commonly introduced to directly add the position information to the context representations. Unlikely, our approach aims to derive the relative position encoding from Equation (1) under some constraints. Next, we show that the derived approach is more interpretable by incorporating relative position information with the rotation of context representations.

3 Proposed approach

In this section, we discuss the proposed rotary position embedding (RoPE). We first formulate the relative position encoding problem in Section (3.1), we then derive the RoPE in Section (3.2) and investigate its properties in Section (3.3).

3.1 Formulation

Transformer-based language modeling usually leverages the position information of individual tokens through a self-attention mechanism. As can be observed in Equation (2), $\mathbf{q}_m^\top \mathbf{k}_n$ typically enables knowledge conveyance between tokens at different positions. In order to incorporate relative position information, we require the inner product of query $\mathbf{q}_m$ and key $\mathbf{k}_n$ to be formulated by a function g , which takes only the word embeddings $\mathbf{x}_m$, $\mathbf{x}_n$, and their relative position $m - n$ as input variables. In other words, we hope that the inner product encodes position information only in the relative form:

$$\langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, m - n). \quad (11)$$

The ultimate goal is to find an equivalent encoding mechanism to solve the functions $f_q(\mathbf{x}_m, m)$ and $f_k(\mathbf{x}_n, n)$ to conform the aforementioned relation.

3.2 Rotary position embedding

3.2.1 A 2D case

We begin with a simple case with a dimension $d = 2$. Under these settings, we make use of the geometric property of vectors on a 2D plane and its complex form to prove (refer Section (3.4.1) for more details) that a solution to our formulation Equation (11) is:

$$\begin{aligned} f_q(\mathbf{x}_m, m) &= (\mathbf{W}_q \mathbf{x}_m) e^{im\theta} \\ f_k(\mathbf{x}_n, n) &= (\mathbf{W}_k \mathbf{x}_n) e^{in\theta} \\ g(\mathbf{x}_m, \mathbf{x}_n, m - n) &= \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}] \end{aligned} \quad (12)$$

where $\text{Re}[\cdot]$ is the real part of a complex number and $(\mathbf{W}_k \mathbf{x}_n)^*$ represents the conjugate complex number of $(\mathbf{W}_k \mathbf{x}_n)$. $\theta \in \mathbb{R}$ is a preset non-zero constant. We can further write $f_{\{q,k\}}$ in a multiplication matrix:

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix} \quad (13)$$

where $(x_m^{(1)}, x_m^{(2)})$ is $\mathbf{x}_m$ expressed in the 2D coordinates. Similarly, g can be viewed as a matrix and thus enables the solution of formulation in Section (3.1) under the 2D case. Specifically, incorporating the relative position embedding is straightforward: simply rotate the affine-transformed word embedding vector by amount of angle multiples of its position index and thus interprets the intuition behind *Rotary Position Embedding*.

He等人 [2020] 认为，两个标记的相对位置只能使用公式(6)的中间两项来完全建模。因此，绝对位置嵌入 $\mathbf{p}_m$ 和 $\mathbf{p}_n$ 被简单地替换为相对位置嵌入 $\tilde{\mathbf{p}}_{m-n}$:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \tilde{\mathbf{p}}_{m-n} + \tilde{\mathbf{p}}_{m-n}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n \quad (10)$$

对Radford和Narasimhan [2018] 提出的四种相对位置嵌入变体的比较表明，与公式(10)相似的变体在其余三种变体中是最有效的。一般来说，所有这些方法都试图基于公式(3)在公式(2)中自注意力设置下的分解来修改公式(6)，该公式最初由Vaswani等人 [2017]提出。它们通常直接将位置信息添加到上下文表示中。不同地，我们的方法旨在在某些约束下从公式(1)推导出相对位置编码。接下来，我们通过结合相对位置信息与上下文表示的旋转，展示推导出的方法更具可解释性。

3 提出方法

在本节中，我们讨论提出的旋转位置嵌入(RoPE)。我们首先在(3.1)部分中提出相对位置编码问题，然后在(3.2)部分中推导出RoPE，并在(3.3)部分中研究其性质。

3.1 公式(2)

基于 Transformer 的语言建模通常通过自注意力机制利用单个词的位置信息。如公式(2)所示， $\mathbf{q}_m^\top \mathbf{k}_n$ 通常使不同位置的词之间能够传递知识。为了结合相对位置信息，我们需要将查询 $\mathbf{q}_m$ 和键 $\mathbf{k}_n$ 的内积通过一个函数 g 来定义，该函数仅以词嵌入 $\mathbf{x}_m$ 、 $\mathbf{x}_n$ 及其相对位置 $m - n$ 作为输入变量。换句话说，我们希望内积仅以相对形式编码位置信息：

$$\langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, m - n). \quad (11)$$

最终目标是找到一个等效的编码机制来解决函数 $f_q(\mathbf{x}_m, m)$ 和 $f_k(\mathbf{x}_n, n)$ ，使其符合上述关系。

3.2 旋转位置嵌入

3.2.1 二维情况

我们从一个维度为 $d = 2$ 的简单情况开始。在这些设置下，我们利用二维平面上的向量的几何属性及其复数形式来证明（参考第(3.4.1)节了解更多细节），我们公式的解 Equation(11) 是：

$$\begin{aligned} f_q(\mathbf{x}_m, m) &= (\mathbf{W}_q \mathbf{x}_m) e^{im\theta} \\ f_k(\mathbf{x}_n, n) &= (\mathbf{W}_k \mathbf{x}_n) e^{in\theta} \\ g(\mathbf{x}_m, \mathbf{x}_n, m - n) &= \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}] \end{aligned} \quad (12)$$

其中 $\text{Re}[\cdot]$ 是复数的实部， $(\mathbf{W}_k \mathbf{x}_n)^*$ 表示 $(\mathbf{W}_k \mathbf{x}_n)$ 的共轭复数。 $\theta \in \mathbb{R}$ 是预设的非零常数。我们可以进一步将 $f_{\{q,k\}}$ 写成乘法矩阵：

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix} \quad (13)$$

在二维坐标中如何表示 $(x_m^{(1)}, x_m^{(2)})$ 。类似地， g 可以被视为一个矩阵，从而使得在二维情况下能够解决第(3.1)节中的公式。具体来说，结合相对位置嵌入是直接的：只需将仿射变换词嵌入向量旋转其位置索引的角度倍数，从而解释旋转位置嵌入背后的直观意义。

3.2.2 General form

In order to generalize our results in 2D to any $\mathbf{x}_i \in \mathbb{R}^d$ where d is even, we divide the d -dimension space into $d/2$ sub-spaces and combine them in the merit of the linearity of the inner product, turning $f_{\{q,k\}}$ into:

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \mathbf{R}_{\Theta, m}^d \mathbf{W}_{\{q,k\}} \mathbf{x}_m \quad (14)$$

where

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix} \quad (15)$$

is the rotary matrix with pre-defined parameters $\Theta = \{\theta_i = 10000^{-2(i-1)/d}, i \in [1, 2, \dots, d/2]\}$. A graphic illustration of RoPE is shown in Figure (1). Applying our RoPE to self-attention in Equation (2), we obtain:

$$\mathbf{q}_m^\top \mathbf{k}_n = (\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = \mathbf{x}^\top \mathbf{W}_q \mathbf{R}_{\Theta, n-m}^d \mathbf{W}_k \mathbf{x}_n \quad (16)$$

where $\mathbf{R}_{\Theta, n-m}^d = (\mathbf{R}_{\Theta, m}^d)^\top \mathbf{R}_{\Theta, n}^d$. Note that $\mathbf{R}_{\Theta}^d$ is an orthogonal matrix, which ensures stability during the process of encoding position information. In addition, due to the sparsity of $\mathbf{R}_{\Theta}^d$, applying matrix multiplication directly as in Equation (16) is not computationally efficient; we provide another realization in theoretical explanation.

In contrast to the additive nature of position embedding method adopted in the previous works, i.e., Equations (3) to (10), our approach is multiplicative. Moreover, RoPE naturally incorporates relative position information through rotation matrix product instead of altering terms in the expanded formulation of additive position encoding when applied with self-attention.

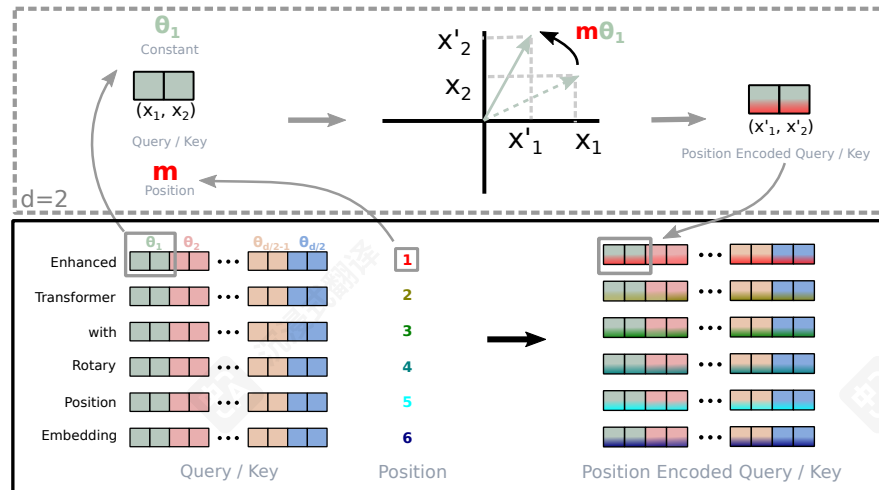


Figure 1: Implementation of Rotary Position Embedding(RoPE).

3.3 Properties of RoPE

Long-term decay: Following Vaswani et al. [2017], we set $\theta_i = 10000^{-2i/d}$. One can prove that this setting provides a long-term decay property (refer to Section (3.4.3) for more details), which means the inner-product will decay when the relative position increase. This property coincides with the intuition that a pair of tokens with a long relative distance should have less connection.

RoPE with linear attention: The self-attention can be rewritten in a more general form.

3.2.2 更一般的形式

为了将我们在二维中的结果推广到任何 $\mathbf{x}_i \in \mathbb{R}^d$ ，其中 d 是偶数，我们将 d -d 维空间划分为 $d/2$ 个子空间，并利用内积的线性特性将它们组合起来，将 $f_{\{q,k\}}$ 转换为：

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \mathbf{R}_{\Theta, m}^d \mathbf{W}_{\{q,k\}} \mathbf{x}_m \quad (14)$$

where

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix} \quad (15)$$

是具有预定义参数的旋转矩阵 $\Theta = \{\theta_i = 10000^{-2(i-1)/d}, i \in [1, 2, \dots, d/2]\}$ 。RoPE 的图形说明如图(1)所示。将我们的 RoPE 应用于公式(2)中的自注意力，我们得到：

$$\mathbf{q}_m^\top \mathbf{k}_n = (\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = \mathbf{x}^\top \mathbf{W}_q \mathbf{R}_{\Theta, n-m}^d \mathbf{W}_k \mathbf{x}_n \quad (16)$$

其中 $\mathbf{R}_{\Theta, n-m}^d = (\mathbf{R}_{\Theta, m}^d)^\top \mathbf{R}_{\Theta, n}^d$ 。请注意， $\mathbf{R}_{\Theta}^d$ 是一个正交矩阵，这确保了在编码位置信息的过程中稳定性。此外，由于 $\mathbf{R}_{\Theta}^d$ 的稀疏性，直接按公式(16)应用矩阵乘法在计算上不是高效的；我们在理论解释中提供了另一种实现方式。

与先前工作中采用的加性位置嵌入方法（即公式(3)至(10)）的加性特性不同，我们的方法是乘法的。此外，RoPE 通过旋转矩阵乘法自然地集成了相对位置信息，而不是在应用自注意力时改变加性位置编码的扩展公式中的项。

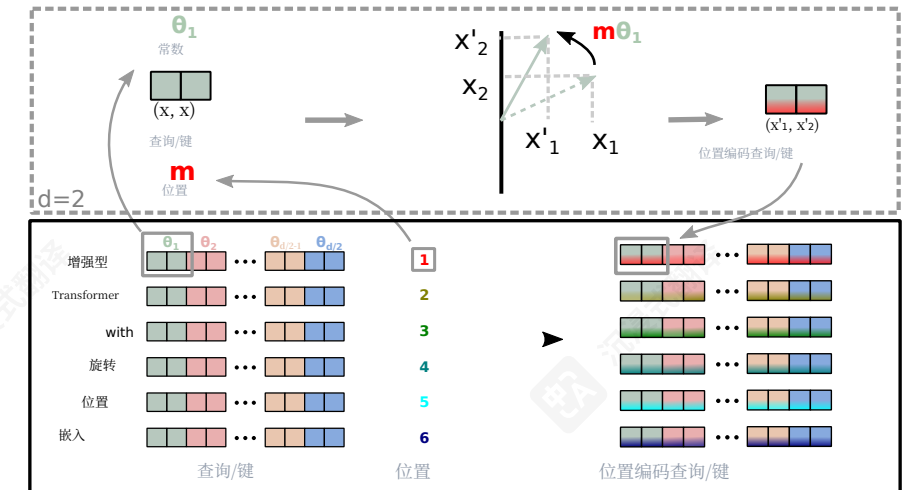


图1: 旋转位置嵌入(RoPE)的实现。

3.3 RoPE的性质

长期衰减: 遵循Vaswani等人 [2017]，我们设置 $\theta_i = 10000^{-2i/d}$ 。可以证明，这种设置具有长期衰减特性（参考第(3.4.3)节了解更多细节），这意味着当相对位置增加时，内积将衰减。这一特性与直观理解一致，即具有较长相对距离的token对应该具有较弱的连接。

RoPE with 线性注意力: 自注意力可以写成更一般的形式。

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n)}. \quad (17)$$

The original self-attention chooses $\text{sim}(\mathbf{q}_m, \mathbf{k}_n) = \exp(\mathbf{q}_m^\top \mathbf{k}_n / \sqrt{d})$. Note that the original self-attention should compute the inner product of query and key for every pair of tokens, which has a quadratic complexity $\mathcal{O}(N^2)$. Follow Katharopoulos et al. [2020], the linear attentions reformulate Equation (17) as

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \phi(\mathbf{q}_m)^\top \varphi(\mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \phi(\mathbf{q}_m)^\top \varphi(\mathbf{k}_n)}, \quad (18)$$

where $\phi(\cdot), \varphi(\cdot)$ are usually non-negative functions. The authors of Katharopoulos et al. [2020] have proposed $\phi(x) = \varphi(x) = \text{elu}(x) + 1$ and first computed the multiplication between keys and values using the associative property of matrix multiplication. A softmax function is used in Shen et al. [2021] to normalize queries and keys separately before the inner product, which is equivalent to $\phi(\mathbf{q}_i) = \text{softmax}(\mathbf{q}_i)$ and $\phi(\mathbf{k}_j) = \exp(\mathbf{k}_j)$. For more details about linear attention, we encourage readers to refer to original papers. In this section, we focus on discussing incorporating RoPE with Equation (18). Since RoPE injects position information by rotation, which keeps the norm of hidden representations unchanged, we can combine RoPE with linear attention by multiplying the rotation matrix with the outputs of the non-negative functions.

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N (\mathbf{R}_{\Theta, m}^d \phi(\mathbf{q}_m))^\top (\mathbf{R}_{\Theta, n}^d \varphi(\mathbf{k}_n)) \mathbf{v}_n}{\sum_{n=1}^N \phi(\mathbf{q}_m)^\top \varphi(\mathbf{k}_n)}. \quad (19)$$

It is noteworthy that we keep the denominator unchanged to avoid the risk of dividing zero, and the summation in the numerator could contain negative terms. Although the weights for each value $\mathbf{v}_i$ in Equation (19) are not strictly probabilistic normalized, we kindly argue that the computation can still model the importance of values.

3.4 Theoretical Explanation

3.4.1 Derivation of RoPE under 2D

Under the case of $d = 2$, we consider two-word embedding vectors $\mathbf{x}_q, \mathbf{x}_k$ corresponds to query and key and their position m and n , respectively. According to eq. (1), their position-encoded counterparts are:

$$\begin{aligned} \mathbf{q}_m &= f_q(\mathbf{x}_q, m), \\ \mathbf{k}_n &= f_k(\mathbf{x}_k, n), \end{aligned} \quad (20)$$

where the subscripts of $\mathbf{q}_m$ and $\mathbf{k}_n$ indicate the encoded positions information. Assume that there exists a function g that defines the inner product between vectors produced by $f_{\{q,k\}}$:

$$\mathbf{q}_m^\top \mathbf{k}_n = \langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, n - m), \quad (21)$$

we further require below initial condition to be satisfied:

$$\begin{aligned} \mathbf{q} &= f_q(\mathbf{x}_q, 0), \\ \mathbf{k} &= f_k(\mathbf{x}_k, 0), \end{aligned} \quad (22)$$

which can be read as the vectors with empty position information encoded. Given these settings, we attempt to find a solution of f_q, f_k . First, we take advantage of the geometric meaning of vector in 2D and its complex counter part, decompose functions in Equations (20) and (21) into:

$$\begin{aligned} f_q(\mathbf{x}_q, m) &= R_q(\mathbf{x}_q, m) e^{i\Theta_q(\mathbf{x}_q, m)}, \\ f_k(\mathbf{x}_k, n) &= R_k(\mathbf{x}_k, n) e^{i\Theta_k(\mathbf{x}_k, n)}, \\ g(\mathbf{x}_q, \mathbf{x}_k, n - m) &= R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) e^{i\Theta_g(\mathbf{x}_q, \mathbf{x}_k, n - m)}, \end{aligned} \quad (23)$$

where R_f, R_g and Θ_f, Θ_g are the radical and angular components for $f_{\{q,k\}}$ and g , respectively. Plug them into Equation (21), we get the relation:

$$\begin{aligned} R_q(\mathbf{x}_q, m) R_k(\mathbf{x}_k, n) &= R_g(\mathbf{x}_q, \mathbf{x}_k, n - m), \\ \Theta_k(\mathbf{x}_k, n) - \Theta_q(\mathbf{x}_q, m) &= \Theta_g(\mathbf{x}_q, \mathbf{x}_k, n - m), \end{aligned} \quad (24)$$

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \text{sim}(\mathbf{q}_m, \mathbf{k}_n)}. \quad (17)$$

原始自注意力选择 $\text{sim}(\mathbf{q}_m, \mathbf{k}_n) = \exp(\mathbf{q}_m^\top \mathbf{k}_n / \sqrt{d})$ 。请注意，原始自注意力应该为每对 token 计算查询和键的内积，这具有二次复杂度 $\mathcal{O}(N^2)$ 。遵循 Katharopoulos 等人 [2020]，线性注意力机制将方程 (17) 改写为

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N \phi(\mathbf{q}_m)^\top \varphi(\mathbf{k}_n) \mathbf{v}_n}{\sum_{n=1}^N \phi(\mathbf{q}_m)^\top \varphi(\mathbf{k}_n)}, \quad (18)$$

其中 $\phi(\cdot), \varphi(\cdot)$ ，通常是 非负函数。Katharopoulos 等人 [2020] 提出了 $\phi(x) = \varphi(x) = \text{elu}(x) + 1$ 并首次使用矩阵乘法的结合律计算了键和值之间的乘法。Shen 等人 [2021] 在内积之前使用 softmax 函数分别对查询和键进行归一化，这相当于 $\phi(\mathbf{q}_i) = \text{softmax}(\mathbf{q}_i)$ 和 $\phi(\mathbf{k}_j) = \exp(\mathbf{k}_j)$ 。有关线性注意力的更多细节，我们建议读者参考原始论文。在本节中，我们专注于讨论将 RoPE 与公式 (18) 结合。由于 RoPE 通过旋转注入位置信息，这保持了隐藏表示的范数不变，因此我们可以通过将旋转矩阵与非负函数的输出相乘，将 RoPE 与线性注意力结合起来。

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_m = \frac{\sum_{n=1}^N (\mathbf{R}_{\Theta, m}^d \phi(\mathbf{q}_m))^\top (\mathbf{R}_{\Theta, n}^d \varphi(\mathbf{k}_n)) \mathbf{v}_n}{\sum_{n=1}^N \phi(\mathbf{q}_m)^\top \varphi(\mathbf{k}_n)}. \quad (19)$$

值得注意的是，我们保持分母不变以避免除零的风险，分子中的求和可能包含负项。尽管公式 (19) 中每个值 $\mathbf{v}_i$ 的权重不是严格概率归一化的，但我们恳切地认为，计算仍然可以模拟值的重要性。

3.4 理论解释

3.4.1 在2D下推导RoPE

在 $d = 2$ 的情况下，我们考虑两个词嵌入向量 $\mathbf{x}_q, \mathbf{x}_k$ 分别对应查询和键，并且它们的位置分别是 m 和 n 。根据等式(1)，它们的位置编码的对应物是：

$$\begin{aligned} \mathbf{q}_m &= f_q(\mathbf{x}_q, m), \\ \mathbf{k}_n &= f_k(\mathbf{x}_k, n), \end{aligned} \quad (20)$$

其中 $\mathbf{q}_m$ 和 $\mathbf{k}_n$ 的下标表示编码的位置信息。假设存在一个函数 g ，它定义了由 $f_{\{q,k\}}$ 产生的向量之间的内积。

$$\mathbf{q}_m^\top \mathbf{k}_n = \langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, n - m), \quad (21)$$

我们进一步要求满足以下初始条件：

$$\begin{aligned} \mathbf{q} &= f_q(\mathbf{x}_q, 0), \\ \mathbf{k} &= f_k(\mathbf{x}_k, 0), \end{aligned} \quad (22)$$

这可以理解为带有空位置信息编码的向量。给定这些设置，我们尝试找到 f_q, f_k 的解。首先，我们利用向量在2D中的几何意义及其复数对应部分，将方程(20)和(21)中的分解函数分解为：

$$\begin{aligned} f_q(\mathbf{x}_q, m) &= R_q(\mathbf{x}_q, m) e^{i\Theta_q(\mathbf{x}_q, m)}, \\ f_k(\mathbf{x}_k, n) &= R_k(\mathbf{x}_k, n) e^{i\Theta_k(\mathbf{x}_k, n)}, \\ g(\mathbf{x}_q, \mathbf{x}_k, n - m) &= R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) e^{i\Theta_g(\mathbf{x}_q, \mathbf{x}_k, n - m)}, \end{aligned} \quad (23)$$

其中 R_f, R_g 和 Θ_f, Θ_g 分别是 $f_{\{q,k\}}$ 和 g 的根式和角分量。将它们代入公式(21)，我们得到关系：

$$\begin{aligned} R_q(\mathbf{x}_q, m) R_k(\mathbf{x}_k, n) &= R_g(\mathbf{x}_q, \mathbf{x}_k, n - m), \\ \Theta_k(\mathbf{x}_k, n) - \Theta_q(\mathbf{x}_q, m) &= \Theta_g(\mathbf{x}_q, \mathbf{x}_k, n - m), \end{aligned} \quad (24)$$

with the corresponding initial condition as:

$$\begin{aligned}\mathbf{q} &= \|\mathbf{q}\|e^{i\theta_q} = R_q(\mathbf{x}_q, 0)e^{i\Theta_q(\mathbf{x}_q, 0)}, \\ \mathbf{k} &= \|\mathbf{k}\|e^{i\theta_k} = R_k(\mathbf{x}_k, 0)e^{i\Theta_k(\mathbf{x}_k, 0)},\end{aligned}\quad (25)$$

where $\|\mathbf{q}\|$, $\|\mathbf{k}\|$ and θ_q, θ_k are the radial and angular part of $\mathbf{q}$ and $\mathbf{k}$ on the 2D plane.

Next, we set $m = n$ in Equation (24) and take into account initial conditions in Equation (25):

$$R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, m) = R_g(\mathbf{x}_q, \mathbf{x}_k, 0) = R_q(\mathbf{x}_q, 0)R_k(\mathbf{x}_k, 0) = \|\mathbf{q}\|\|\mathbf{k}\|, \quad (26a)$$

$$\Theta_k(\mathbf{x}_k, m) - \Theta_q(\mathbf{x}_q, m) = \Theta_g(\mathbf{x}_q, \mathbf{x}_k, 0) = \Theta_k(\mathbf{x}_k, 0) - \Theta_q(\mathbf{x}_q, 0) = \theta_k - \theta_q. \quad (26b)$$

On one hand, from, a straightforward solution of R_f could be formed from Equation (26a) :

$$\begin{aligned}R_q(\mathbf{x}_q, m) &= R_q(\mathbf{x}_q, 0) = \|\mathbf{q}\| \\ R_k(\mathbf{x}_k, n) &= R_k(\mathbf{x}_k, 0) = \|\mathbf{k}\| \\ R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) &= R_g(\mathbf{x}_q, \mathbf{x}_k, 0) = \|\mathbf{q}\|\|\mathbf{k}\|\end{aligned}\quad (27)$$

which interprets the radial functions R_q, R_k and R_g are independent from the position information. On the other hand, as can be noticed in Equation (26b), $\Theta_q(\mathbf{x}_q, m) - \theta_q = \Theta_k(\mathbf{x}_k, m) - \theta_k$ indicates that the angular functions does not dependent on query and key, we set them to $\Theta_f := \Theta_q = \Theta_k$ and term $\Theta_f(\mathbf{x}_{\{q,k\}}, m) - \theta_{\{q,k\}}$ is a function of position m and is independent of word embedding $\mathbf{x}_{\{q,k\}}$, we denote it as $\phi(m)$, yielding:

$$\Theta_f(\mathbf{x}_{\{q,k\}}, m) = \phi(m) + \theta_{\{q,k\}}, \quad (28)$$

Further, by plugging $n = m + 1$ to Equation (24) and consider the above equation, we can get:

$$\phi(m + 1) - \phi(m) = \Theta_g(\mathbf{x}_q, \mathbf{x}_k, 1) + \theta_q - \theta_k, \quad (29)$$

Since RHS is a constant irrelevant to m , $\phi(m)$ with continuous integer inputs produce an arithmetic progression:

$$\phi(m) = m\theta + \gamma, \quad (30)$$

where $\theta, \gamma \in \mathbb{R}$ are constants and θ is non-zero. To summarize our solutions from Equations (27) to (30):

$$\begin{aligned}f_q(\mathbf{x}_q, m) &= \|\mathbf{q}\|e^{i\theta_q + m\theta + \gamma} = \mathbf{q}e^{i(m\theta + \gamma)}, \\ f_k(\mathbf{x}_k, n) &= \|\mathbf{k}\|e^{i\theta_k + n\theta + \gamma} = \mathbf{k}e^{i(n\theta + \gamma)}.\end{aligned}\quad (31)$$

Note that we do not apply any constrains to f_q and f_k of Equation (22), thus $f_q(\mathbf{x}_m, 0)$ and $f_k(\mathbf{x}_n, 0)$ are left to choose freely. To make our results comparable to Equation (3), we define:

$$\begin{aligned}\mathbf{q} &= f_q(\mathbf{x}_m, 0) = \mathbf{W}_q\mathbf{x}_m, \\ \mathbf{k} &= f_k(\mathbf{x}_n, 0) = \mathbf{W}_k\mathbf{x}_n.\end{aligned}\quad (32)$$

Then, we simply set $\gamma = 0$ in Equation (31) of the final solution:

$$\begin{aligned}f_q(\mathbf{x}_m, m) &= (\mathbf{W}_q\mathbf{x}_m)e^{im\theta}, \\ f_k(\mathbf{x}_n, n) &= (\mathbf{W}_k\mathbf{x}_n)e^{in\theta}.\end{aligned}\quad (33)$$

3.4.2 Computational efficient realization of rotary matrix multiplication

Taking the advantage of the sparsity of $\mathbf{R}_{\Theta, m}^d$ in Equation (15), a more computational efficient realization of a multiplication of $\mathbf{R}_{\Theta}^d$ and $\mathbf{x} \in \mathbb{R}^d$ is:

$$\mathbf{R}_{\Theta, m}^d \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \cos m\theta_2 \\ \cos m\theta_2 \\ \vdots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_1 \\ -x_4 \\ x_3 \\ \vdots \\ -x_d \\ x_{d-1} \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \sin m\theta_2 \\ \sin m\theta_2 \\ \vdots \\ \sin m\theta_{d/2} \\ \sin m\theta_{d/2} \end{pmatrix} \quad (34)$$

其相应的初始条件为:

$$\begin{aligned}\mathbf{q} &= \|\mathbf{q}\|e^{i\theta_q} = R_q(\mathbf{x}_q, 0)e^{i\Theta_q(\mathbf{x}_q, 0)}, \\ \mathbf{k} &= \|\mathbf{k}\|e^{i\theta_k} = R_k(\mathbf{x}_k, 0)e^{i\Theta_k(\mathbf{x}_k, 0)},\end{aligned}\quad (25)$$

其中 $\|\mathbf{q}\|$, $\|\mathbf{k}\|$ 和 θ_q, θ_k 是二维平面上的 $\mathbf{q}$ 和 $\mathbf{k}$ 的径向和角部分。

接下来, 我们在公式 (24) 中设置 $m = n$, 并考虑公式 (25) 中的初始条件:

$$R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, m) = R_g(\mathbf{x}_q, \mathbf{x}_k, 0) = R_q(\mathbf{x}_q, 0)R_k(\mathbf{x}_k, 0) = \|\mathbf{q}\|\|\mathbf{k}\|, \quad (26a)$$

$$\Theta_k(\mathbf{x}_k, m) - \Theta_q(\mathbf{x}_q, m) = \Theta_g(\mathbf{x}_q, \mathbf{x}_k, 0) = \Theta_k(\mathbf{x}_k, 0) - \Theta_q(\mathbf{x}_q, 0) = \theta_k - \theta_q. \quad (26b)$$

一方面, 从, R_f 的直接解可以由公式 (26a) 形成:

$$\begin{aligned}R_q(\mathbf{x}_q, m) &= R_q(\mathbf{x}_q, 0) = \|\mathbf{q}\| \\ R_k(\mathbf{x}_k, n) &= R_k(\mathbf{x}_k, 0) = \|\mathbf{k}\| \\ R_g(\mathbf{x}_q, \mathbf{x}_k, n - m) &= R_g(\mathbf{x}_q, \mathbf{x}_k, 0) = \|\mathbf{q}\|\|\mathbf{k}\|\end{aligned}\quad (27)$$

该函数解释径向函数 R_q, R_k 和 R_g 与位置信息独立。另一方面, 如公式 (26b) 所示,

$\Theta_q(\mathbf{x}_q, m) - \theta_q = \Theta_k(\mathbf{x}_k, m) - \theta_k$ 表明角函数不依赖于查询和键, 我们将其设置为 $\Theta_f := \Theta_q = \Theta_k$, 项 $\Theta_f(\mathbf{x}_{\{q,k\}}, m) - \theta_{\{q,k\}}$ 是位置 m 的函数, 且与词嵌入 $\mathbf{x}_{\{q,k\}}$ 独立, 我们将其表示为 $\phi(m)$, 得到:

$$\Theta_f(\mathbf{x}_{\{q,k\}}, m) = \phi(m) + \theta_{\{q,k\}}, \quad (28)$$

此外, 通过将 $n = m + 1$ 代入公式 (24) 并考虑上述公式, 我们可以得到:

$$\phi(m + 1) - \phi(m) = \Theta_g(\mathbf{x}_q, \mathbf{x}_k, 1) + \theta_q - \theta_k, \quad (29)$$

由于 RHS 是一个与 m 、 $\phi(m)$ 无关的常数, 连续整数输入产生一个算术级数:

$$\phi(m) = m\theta + \gamma, \quad (30)$$

其中 $\theta, \gamma \in \mathbb{R}$ 是常数, θ 不为零。总结我们从公式(27)至(30)的解决方案:

$$\begin{aligned}f_q(\mathbf{x}_q, m) &= \|\mathbf{q}\|e^{i\theta_q + m\theta + \gamma} = \mathbf{q}e^{i(m\theta + \gamma)}, \\ f_k(\mathbf{x}_k, n) &= \|\mathbf{k}\|e^{i\theta_k + n\theta + \gamma} = \mathbf{k}e^{i(n\theta + \gamma)}.\end{aligned}\quad (31)$$

注意, 我们没有对公式(22)的 f_q 和 f_k 施加任何约束, 因此 $f_q(\mathbf{x}_m, 0)$ 和 $f_k(\mathbf{x}_n, 0)$ 可以自由选择。为了使我们的结果与公式(3)可比, 我们定义:

$$\begin{aligned}\mathbf{q} &= f_q(\mathbf{x}_m, 0) = \mathbf{W}_q\mathbf{x}_m, \\ \mathbf{k} &= f_k(\mathbf{x}_n, 0) = \mathbf{W}_k\mathbf{x}_n.\end{aligned}\quad (32)$$

然后, 我们在最终解的公式(31)中简单地设置 $\gamma = 0$:

$$\begin{aligned}f_q(\mathbf{x}_m, m) &= (\mathbf{W}_q\mathbf{x}_m)e^{im\theta}, \\ f_k(\mathbf{x}_n, n) &= (\mathbf{W}_k\mathbf{x}_n)e^{in\theta}.\end{aligned}\quad (33)$$

3.4.2 计算高效的旋转矩阵乘法实现

利用 $\mathbf{R}_{\Theta, m}^d$ 的稀疏性, 在公式(15)中, $\mathbf{R}_{\Theta}^d$ 和 $\mathbf{x} \in \mathbb{R}^d$ 的乘法的一种计算更高效的实现是:

$$\mathbf{R}_{\Theta, m}^d \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ \vdots \\ x_{d-1} \\ x_d \end{pmatrix} \otimes \begin{pmatrix} \cos m\theta_1 \\ \cos m\theta_1 \\ \cos m\theta_2 \\ \cos m\theta_2 \\ \vdots \\ \cos m\theta_{d/2} \\ \cos m\theta_{d/2} \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_1 \\ -x_4 \\ x_3 \\ \vdots \\ -x_d \\ x_{d-1} \end{pmatrix} \otimes \begin{pmatrix} \sin m\theta_1 \\ \sin m\theta_1 \\ \sin m\theta_2 \\ \sin m\theta_2 \\ \vdots \\ \sin m\theta_{d/2} \\ \sin m\theta_{d/2} \end{pmatrix} \quad (34)$$

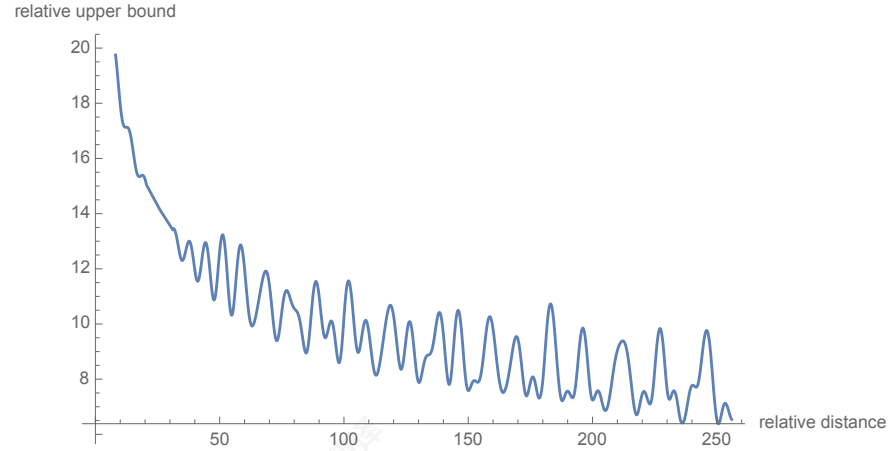


Figure 2: Long-term decay of RoPE.

3.4.3 Long-term decay of RoPE

We can group entries of vectors $\mathbf{q} = \mathbf{W}_q \mathbf{x}_m$ and $\mathbf{k} = \mathbf{W}_k \mathbf{x}_n$ in pairs, and the inner product of RoPE in Equation (16) can be written as a complex number multiplication.

$$(\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = \text{Re} \left[\sum_{i=0}^{d/2-1} \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^* e^{i(m-n)\theta_i} \right] \quad (35)$$

where $\mathbf{q}_{[2i:2i+1]}$ represents the $2i^{th}$ to $(2i+1)^{th}$ entries of $\mathbf{q}$. Denote $h_i = \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^*$ and $S_j = \sum_{i=0}^{j-1} e^{i(m-n)\theta_i}$, and let $h_{d/2} = 0$ and $S_0 = 0$, we can rewrite the summation using Abel transformation

$$\sum_{i=0}^{d/2-1} \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^* e^{i(m-n)\theta_i} = \sum_{i=0}^{d/2-1} h_i (S_{i+1} - S_i) = - \sum_{i=0}^{d/2-1} S_{i+1} (h_{i+1} - h_i). \quad (36)$$

Thus,

$$\begin{aligned} \left| \sum_{i=0}^{d/2-1} \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^* e^{i(m-n)\theta_i} \right| &= \left| \sum_{i=0}^{d/2-1} S_{i+1} (h_{i+1} - h_i) \right| \\ &\leq \sum_{i=0}^{d/2-1} |S_{i+1}| |h_{i+1} - h_i| \\ &\leq \left(\max_i |h_{i+1} - h_i| \right) \sum_{i=0}^{d/2-1} |S_{i+1}| \end{aligned} \quad (37)$$

Note that the value of $\frac{1}{d/2} \sum_{i=1}^{d/2} |S_i|$ decay with the relative distance $m - n$ increases by setting $\theta_i = 10000^{-2i/d}$, as shown in Figure (2).

4 Experiments and Evaluation

We evaluate the proposed RoFormer on various NLP tasks as follows. We validate the performance of the proposed solution on machine translation task Section (4.1). Then, we compare our RoPE implementation with BERTDevlin et al. [2019] during the pre-training stage in Section (4.2). Based on the pre-trained model, in Section (4.3), we further carry out evaluations across different downstream tasks from GLUE benchmarks Singh et al. [2018]. In Addition, we conduct experiments using the proposed RoPE with the linear attention of PerFormer Choromanski et al. [2020] in

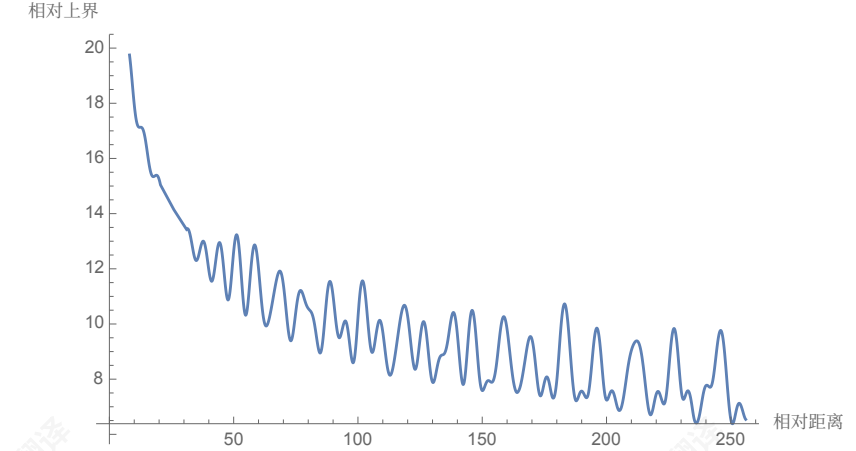


图2: RoPE的长期衰减。

3.4.3 RoPE的长期衰减

我们可以将向量 $\mathbf{q} = \mathbf{W}_q \mathbf{x}_m$ 和 $\mathbf{k} = \mathbf{W}_k \mathbf{x}_n$ 的项成对分组，公式(16)中 RoPE 的内积可以写成一个复数乘法。

$$(\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = \text{Re} \left[\sum_{i=0}^{d/2-1} \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^* e^{i(m-n)\theta_i} \right] \quad (35)$$

其中 $\mathbf{q}_{[2i:2i+1]}$ 表示 $\mathbf{q}$ 的 $2i^{th}$ 到 $2i+1^{th}$ 项。记 $h_i = \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^*$ 和 $S_j = \sum_{i=0}^{j-1} e^{i(m-n)\theta_i}$ ，并令 $h_{d/2} = 0$ 和 $S_0 = 0$ ，我们可以使用阿贝尔变换重写求和

$$\sum_{i=0}^{d/2-1} \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^* e^{i(m-n)\theta_i} = \sum_{i=0}^{d/2-1} h_i (S_{i+1} - S_i) = - \sum_{i=0}^{d/2-1} S_{i+1} (h_{i+1} - h_i). \quad (36)$$

因此，

$$\begin{aligned} \left| \sum_{i=0}^{d/2-1} \mathbf{q}_{[2i:2i+1]} \mathbf{k}_{[2i:2i+1]}^* e^{i(m-n)\theta_i} \right| &= \left| \sum_{i=0}^{d/2-1} S_{i+1} (h_{i+1} - h_i) \right| \\ &\leq \sum_{i=0}^{d/2-1} |S_{i+1}| |h_{i+1} - h_i| \\ &\leq \left(\max_i |h_{i+1} - h_i| \right) \sum_{i=0}^{d/2-1} |S_{i+1}| \end{aligned} \quad (37)$$

请注意， $\frac{1}{d/2} \sum_{i=1}^{d/2} |S_i|$ 衰减随着相对距离 $m - n$ 的增加而通过设置 $\theta_i = 10000^{-2i/d}$ 而增大，如图(2)所示。

4 实验与评估

我们评估了proposed RoFormer在各种NLP任务上的表现。我们验证了proposed解决方案在机器翻译任务 Section (4.1)上的性能。然后，我们在Section (4.2)中的预训练阶段比较了我们的RoPE实现与BERTDevlin等人 [2019]。基于预训练模型，在Section (4.3)中，我们从GLUE基准Singh等人 [2018]的不同下游任务进一步进行评估。此外，我们使用proposed RoPE与PerFormer Choromanski等人 [2020] 的线性注意力进行了实验

Table 1: The proposed RoFormer gives better BLEU scores compared to its baseline alternative Vaswani et al. [2017] on the WMT 2014 English-to-German translation taskBojar et al. [2014].

Model	BLEU
Transformer-baseVaswani et al. [2017]	27.3
RoFormer	27.5

Section (4.4). By the end, additional tests on Chinese data are included in Section (4.5). All the experiments were run on two cloud severs with 4 x V100 GPUs.

4.1 Machine Translation

We first demonstrate the performance of RoFormer on sequence-to-sequence language translation tasks.

4.1.1 Experimental Settings

We choose the standard WMT 2014 English-German datasetBojar et al. [2014], which consists of approximately 4.5 million sentence pairs. We compare to the transformer-based baseline alternative Vaswani et al. [2017].

4.1.2 Implementation details

We carry out some modifications on self-attention layer of the baseline model Vaswani et al. [2017] to enable RoPE to its learning process. We replicate the setup for English-to-German translation with a vocabulary of 37k based on a joint source and target byte pair encoding(BPE)Sennrich et al. [2015]. During the evaluation, a single model is obtained by averaging the last 5 checkpoints. The result uses beam search with a beam size of 4 and length penalty 0.6. We implement the experiment in PyTorch in the fairseq toolkit (MIT License)Ott et al. [2019]. Our model is optimized with the Adam optimizer using $\beta_1 = 0.9$, $\beta_2 = 0.98$, learning rate is increased linearly from $1e-7$ to $5e-4$ and then decayed proportionally to the inverse square root of the step number. Label smoothing with 0.1 is also adopted. We report the BLEUPapineni et al. [2002] score on the test set as the final metric.

4.1.3 Results

We train the baseline model and our RoFormer under the same settings and report the results in Table (1). As can be seen, our model gives better BLEU scores compared to the baseline Transformer.

4.2 Pre-training Language Modeling

The second experiment is to validate the performance of our proposal in terms of learning contextual representations. To achieve this, we replace the original sinusoidal position encoding of BERT with our RoPE during the pre-training step.

4.2.1 Experimental Settings

We use the BookCorpus Zhu et al. [2015] and the Wikipedia Corpus Foundation [2021] from Huggingface Datasets library (Apache License 2.0) for pre-training. The corpus is further split into train and validation sets at 8:2 ratio. We use the masked language-modeling (MLM) loss values of the training process as an evaluation metric. The well-known BERT Devlin et al. [2019] is adopted as our baseline model. Note that we use bert-base-uncased in our experiments.

4.2.2 Implementation details

For RoFormer, we replace the sinusoidal position encoding in the self-attention block of the baseline model with our proposed RoPE and realizes self-attention according to Equation (16). We train both BERT and RoFormer with batch size 64 and maximum sequence length of 512 for 100k steps. AdamW Loshchilov and Hutter [2017] is used as the optimizer with learning rate $1e-5$.

4.2.3 Results

The MLM loss during pre-training is shown on the left plot of Figure (3). Compare to the vanilla BERT, RoFormer experiences faster convergence.

表1: 所提出的RoFormer与基线替代方案Vaswani等人 [2017]相比, 在WMT 2014英德翻译任务上获得了更好的 BLEU分数Bojar等人 [2014].

模型BLEU	
Transformer基础Vaswani等人 [2017]	27.3
RoFormer	27.5

Section (4.4)。最后, 在Section(4.5)中包含了额外的中文数据测试。所有实验都在两台云服务器上运行, 配置为4 x V100 GPUs。

4.1 机器翻译

我们首先展示了RoFormer在序列到序列语言翻译任务上的性能。

4.1.1 实验设置

我们选择了标准的WMT 2014英德数据集Bojar等人 [2014], 该数据集包含约450万个句子对。我们将RoFormer与基于Transformer的基线替代方案Vaswani等人 [2017]进行比较。

4.1.2 实现细节

我们对基线模型Vaswani等人 [2017] 的自注意力层进行了一些修改, 以使RoPE能够参与其学习过程。我们基于联合源语言和目标语言的字节对编码(BPE)Sennrich等人 [2015], 使用3.7k词汇量复制了英德翻译的设置。在评估期间, 通过平均最后5个检查点获得单个模型。结果使用束大小为4、长度惩罚为0.6的束搜索。我们在fairseq工具包 (MIT许可证) Ott等人 [2019]中用PyTorch实现了该实验。我们的模型使用Adam优化器进行优化, 使用 $\beta_1 = 0.9$, $\beta_2 = 0.98$, 学习率从 $1e-7$ 线性增加到 $5e-4$, 然后按步数的平方根倒数比例衰减。我们还采用了0.1的标签平滑。我们在测试集上报告了Papineni等人 [2002] 的BLEU分数作为最终指标。

4.1.3 结果

我们在相同设置下训练基线模型和我们的 RoFormer, 并在表(1)中报告结果。如所见, 我们的模型在 BLEU 分数方面优于基线 Transformer。

4.2 预训练语言建模

第二个实验是验证我们的提案在学习上下文表示方面的性能。为此, 我们在预训练步骤中将 BERT 的原始正弦位置编码替换为我们的 RoPE。

4.2.1 实验设置

我们使用 Huggingface 数据集库 (Apache 许可证 2.0) 中的 BookCorpus Zhu et al. [2015] 和维基百科语料库 Foundation [2021] 进行预训练。语料库进一步按 8:2 的比例分割为训练集和验证集。我们使用训练过程中的掩码语言模型(MLM)损失值作为评估指标。著名的 BERT Devlin 等人 [2019]被用作我们的基线模型。请注意, 我们在实验中使用 bert-base-uncased。

4.2.2 实现细节

对于 RoFormer, 我们将基线模型中自注意力模块的正弦位置编码替换为我们的 RoPE, 并根据公式 (16) 实现自注意力。我们使用批大小 64 和最大序列长度 512 对 BERT 和 RoFormer 进行 100k 步的训练。AdamW Loshchilov 和 Hutter [2017] 作为优化器, 学习率为 $1e-5$ 。

4.2.3 结果

预训练期间的MLM损失显示在图(3)的左图中。与vanilla BERT相比, RoFormer经历了更快的收敛。

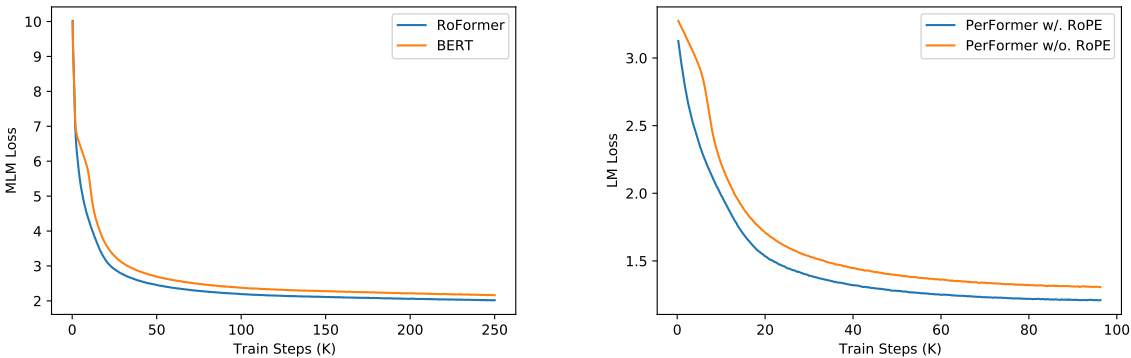


Figure 3: Evaluation of RoPE in language modeling pre-training. **Left:** training loss for BERT and RoFormer. **Right:** training loss for PerFormer with and without RoPE.

4.3 Fine-tuning on GLUE tasks

Consistent with the previous experiments, we fine-tune the weights of our pre-trained RoFormer across various GLUE tasks in order to evaluate its generalization ability on the downstream NLP tasks.

4.3.1 Experimental Settings

We look at several datasets from GLUE, i.e. MRPC Dolan and Brockett [2005], SST-2 Socher et al. [2013], QNLI Rajpurkar et al. [2016], STS-B Al-Natsheh [2017], QQP Chen et al. [2018b] and MNLI Williams et al. [2018]. We use F1-score for MRPC and QQP dataset, spearman correlation for STS-B, and accuracy for the remaining as the evaluation metrics.

4.3.2 Implementation details

We use Huggingface Transformers library (Apache License 2.0) Wolf et al. [2020] to fine-tune each of the aforementioned downstream tasks for 3 epochs, with a maximum sequence length of 512, batch size of 32 and learning rates 2,3,4,5e-5. Following Devlin et al. [2019], we report the best-averaged results on the validation set.

Table 2: Comparing RoFormer and BERT by fine tuning on downstream GLEU tasks.

Model	MRPC	SST-2	QNLI	STS-B	QQP	MNLI(m/mm)
BERTDevlin et al. [2019]	88.9	93.5	90.5	85.8	71.2	84.6/83.4
RoFormer	89.5	90.7	88.0	87.0	86.4	80.2/79.8

4.3.3 Results

The evaluation results of the fine-tuning tasks are reported in Table (2). As can be seen, RoFormer can significantly outperform BERT in three out of six datasets, and the improvements are considerable.

4.4 Performer with RoPE

Performer Choromanski et al. [2020] introduces an alternative attention mechanism, linear attention, which is designed to avoid quadratic computation cost that scales with input sequence length. As discussed in Section (3.3), the proposed RoPE can be easily implemented in the Performer model to realize the relative position encoding while keeping its linearly scaled complexity in self-attention. We demonstrate its performance with the pre-training task of language modeling.

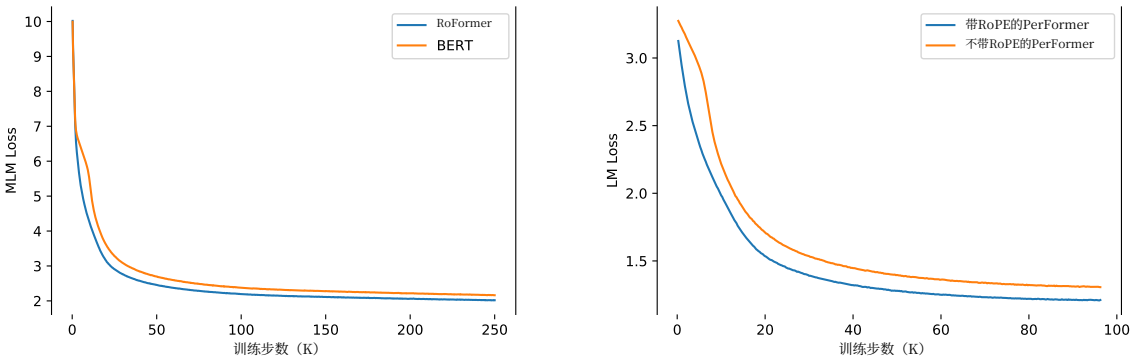


图3：在语言建模预训练中评估RoPE。左：BERT和RoFormer的trainingloss。右：带有和不带有RoPE的PerFormer的训练损失。

4.3 在GLUE任务上微调

与之前的实验一致，我们微调了我们的预训练RoFormer的权重，以在多个GLUE任务上评估其在下游NLP任务上的泛化能力。

4.3.1 实验设置

我们来看GLUE中的几个数据集，即MRPC Dolan和Brockett [2005], SST-2 Socher等人 [2013], QNLI Rajpurkar等人 [2016], STS-B Al-Natsheh [2017],QQP Chen等人 [2018b]以及MNLI Williams等人 [2018]。我们使用F1分数作为MRPC和QQP数据集的评估指标，使用斯皮尔曼相关系数作为STS-B的评估指标，并使用准确率作为其余数据集的评估指标。

4.3.2 实现细节

我们使用 Huggingface 转换器库（Apache 许可证 2.0）Wolf 等人 [2020] 对上述下游任务中的每一个进行微调，共进行 3 个 epoch，最大序列长度为 512，批大小为 32，学习率为 2,3,4,5e-5。遵循 Devlin 等人 [2019]，我们在验证集上报告了最佳平均结果。

表 2：通过在下游 GLEU 任务上进行微调比较 RoFormer 和 BERT。

模型	MRPC	SST-2	QNLI	STS-B	QQP	MNLI(m/mm)
BERTDevlin等人 [2019]	88.9	93.5	90.5	85.8	71.2	84.6/83.4
RoFormer	89.5	90.7	88.0	87.0	86.4	80.2/79.8

4.3.3 结果

微调任务的评估结果报告在表 (2) 中。可以看出，RoFormer 在六个数据集上的三个上可以显著优于 BERT，并且改进是相当大的。

4.4 带有 RoPE 的 Performer

Performer Choromanski 等人 [2020] 提出了一种替代性注意力机制——线性注意力，旨在避免与输入序列长度成二次关系的计算成本。如第(3.3)节所述，所提出的RoPE可以轻松地在PerFormer模型中实现，以实现相对位置编码，同时保持其自注意力中线性缩放的复杂度。我们通过语言建模的预训练任务展示了其性能。

4.4.1 Implementation details

We carry out tests on the Enwik8 dataset Mahoney [2006], which is from English Wikipedia that includes markup, special characters and text in other languages in addition to English text. We incorporate RoPE into the 12 layer char-based PerFormer with 768 dimensions and 12 heads². To better illustrate the efficacy of RoPE, we report the loss curves of the pre-training process with and without RoPE under the same settings, i.e., learning rate 1e-4, batch size 128 and a fixed maximum sequence length of 1024, etc.

4.4.2 Results

As shown on the right plot of Figure (3), substituting RoPE into Performer leads to rapid convergence and lower loss under the same amount of training steps. These improvements, in addition to the linear complexity, make Performer more attractive.

4.5 Evaluation on Chinese Data

In addition to experiments on English data, we show additional results on Chinese data. To validate the performance of RoFormer on long texts, we conduct experiments on long documents whose length exceeds 512 characters.

4.5.1 Implementation

In these experiments, we carried out some modifications on WoBERT Su [2020] by replacing the absolute position embedding with our proposed RoPE. As a cross-comparison with other pre-trained Transformer-based models in Chinese, i.e. BERT Devlin et al. [2019], WoBERT Su [2020], and NEZHA Wei et al. [2019], we tabulate their tokenization level and position embedding information in Table (3).

Table 3: Cross-comparison between our RoFormer and other pre-trained models on Chinese data. 'abs' and 'rel' annotates absolute position embedding and relative position embedding, respectively.

Model	BERTDevlin et al. [2019]	WoBERTSu [2020]	NEZHAWei et al. [2019]	RoFormer
Tokenization level	char	word	char	word
Position embedding	abs.	abs.	rel.	RoPE

4.5.2 Pre-training

We pre-train RoFormer on approximately 34GB of data collected from Chinese Wikipedia, news and forums. The pre-training is carried out in multiple stages with changing batch size and maximum input sequence length in order to adapt the model to various scenarios. As shown in Table (4), the accuracy of RoFormer elevates with an increasing upper bound of sequence length, which demonstrates the ability of RoFormer in dealing with long texts. We claim that this is the attribute to the excellent generalizability of the proposed RoPE.

Table 4: Pre-training strategy of RoFormer on Chinese dataset. The training procedure is divided into various consecutive stages. In each stage, we train the model with a specific combination of maximum sequence length and batch size.

Stage	Max seq length	Batch size	Training steps	Loss	Accuracy
1	512	256	200k	1.73	65.0%
2	1536	256	12.5k	1.61	66.8%
3	256	256	120k	1.75	64.6%
4	128	512	80k	1.83	63.4%
5	1536	256	10k	1.58	67.4%
6	512	512	30k	1.66	66.2%

4.5.3 Downstream Tasks & Dataset

We choose Chinese AI and Law 2019 Similar Case Matching (CAIL2019-SCM)Xiao et al. [2019] dataset to illustrate the ability of RoFormer in dealing with long texts, i.e., semantic text matching. CAIL2019-SCM contains 8964 triplets

²For this experiment, we adopt code (MIT License) from <https://github.com/lucidrains/performer-pytorch>

4.4.1 实现细节

我们在 Enwik8 数据集 Mahoney [2006], 上进行测试, 该数据集来自英语维基百科, 除了英文文本外还包括标记、特殊字符和其他语言的文本。我们将 RoPE 整合到具有 768 维和 12 个头的 12 层基于字符的 PerFormer 中²。为了更好地说明 RoPE 的效果, 我们在相同设置下报告了预训练过程中使用和未使用 RoPE 的损失曲线, 即学习率为 1e-4、批大小为 128 以及固定的最大序列长度为 1024 等。

4.4.2 结果

如图 (3) 右侧所示, 将 RoPE 替换到 Performer 中可以在相同的训练步数下实现快速收敛和更低的损失。除了线性复杂度之外, 这些改进使得 Performer 更具吸引力。

4.5 中文数据评估

除了在英文数据上的实验, 我们还展示了在中文数据上的额外结果。为了验证 RoFormer 在长文本上的性能, 我们在长度超过 512 个字符的长文档上进行了实验。

4.5.1 实现

在这些实验中, 我们对 WoBERT Su [2020] 进行了一些修改, 通过将绝对位置嵌入替换为我们的 RoPE 来实现。与其他中文预训练 Transformer 基础模型 (即 BERT Devlin等人 [2019], WoBERT Su [2020], 和 NEZHA Wei等人 [2019],) 进行交叉比较, 我们在表(3)中列出了它们的分词级别和位置嵌入信息。

表 3: 我们的 RoFormer 与其他中文数据预训练模型之间的交叉比较。'abs' 和 'rel' 分别标注绝对位置嵌入和相对位置嵌入。

模型	BERTDevlin等人 [2019]	WoBERTSu [2020]	NEZHAWei 等人 [2019]	RoFormer
分词级别	char	word	char	word
位置嵌入	abs.	abs.	rel.	RoPE

4.5.2 预训练

我们在中文维基百科、新闻和论坛收集的约 34GB 数据上预训练 RoFormer。预训练分多个阶段进行, 批大小和最大输入序列长度不断变化, 以使模型适应各种场景。如表(4)所示, 随着序列长度上限的增加, RoFormer 的准确率提升, 这展示了 RoFormer 处理长文本的能力。我们声称, 这正是 RoPE 所提出的 RoFormer 优异泛化性的属性。

表4: RoFormer在中文数据集上的预训练策略。训练过程被分为多个连续阶段。在每个阶段中, 我们使用特定组合的最大序列长度和批大小来训练模型。

阶段	最大序列长度	批大小	训练步骤	Loss	准确率
1	512	256	200k	1.73	65.0%
2	1536	256	12.5k	1.61	66.8%
3	256	256	120k	1.75	64.6%
4	128	512	80k	1.83	63.4%
5	1536	256	10k	1.58	67.4%
6	512	512	30k	1.66	66.2%

4.5.3 下游任务与数据集

We choose Chinese AI and Law 2019 Similar Case Matching (CAIL2019-SCM)Xiao et al. [2019] dataset to illustrate the ability of RoFormer in dealing with long texts, i.e., semantic text matching. CAIL2019-SCM contains 8964 triplets

²对于此实验, 我们采用代码 (MIT许可证) 来自 <https://github.com/lucidrains/performer-pytorch>

of cases published by the Supreme People’s Court of China. The input triplet, denoted as (A, B and C), are fact descriptions of three cases. The task is to predict whether the pair (A, B) is closer than (A, C) under a predefined similarity measure. Note that existing methods mostly cannot perform significantly on CAIL2019-SCM dataset due to the length of documents (i.e., mostly more than 512 characters). We split train, validation and test sets based on the well-known ratio 6:2:2.

4.5.4 Results

We apply the pre-trained RoFormer model to CAIL2019-SCM with different input lengths. The model is compared with the pre-trained BERT and WoBERT model on the same pre-training data, as shown in Table (5). With short text cut-offs, i.e., 512, the result from RoFormer is comparable to WoBERT and is slightly better than the BERT implementation. However, when increasing the maximum input text length to 1024, RoFormer outperforms WoBERT by an absolute improvement of 1.5%.

Table 5: Experiment results on CAIL2019-SCM task. Numbers in the first column denote the maximum cut-off sequence length. The results are presented in terms of percent accuracy.

Model	Validation	Test
BERT-512	64.13%	67.77%
WoBERT-512	64.07%	68.10%
RoFormer-512	64.13%	68.29%
RoFormer-1024	66.07%	69.79%

4.5.5 Limitations of the work

Although we provide theoretical groundings as well as promising experimental justifications, our method is limited by following facts:

- Despite the fact that we mathematically format the relative position relations as rotations under 2D sub-spaces, there lacks of thorough explanations on why it converges faster than baseline models that incorporates other position encoding strategies.
- Although we have proved that our model has favourable property of long-term decay for intern-token products, Section (3.3), which is similar to the existing position encoding mechanisms, our model shows superior performance on long texts than peer models, we have not come up with a faithful explanation.

Our proposed RoFormer is built upon the Transformer-based infrastructure, which requires hardware resources for pre-training purpose.

5 Conclusions

In this work, we proposed a new position embedding method that incorporates explicit relative position dependency in self-attention to enhance the performance of transformer architectures. Our theoretical analysis indicates that relative position can be naturally formulated using vector production in self-attention, with absolution position information being encoded through a rotation matrix. In addition, we mathematically illustrated the advantageous properties of the proposed method when applied to the Transformer. Finally, experiments on both English and Chinese benchmark datasets demonstrate that our method encourages faster convergence in pre-training. The experimental results also show that our proposed RoFormer can achieve better performance on long texts task.

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中国最高人民法院发布的案例。输入三元组，记作 (A, B和C)， 是三个案例的事实描述。任务是预测在预定义的相似性度量下， (A, B) 是否比 (A, C) 更接近。请注意，现有方法由于文档长度（即，大多超过512个字符）的原因，在CAIL2019-SCM数据集上表现大多不明显。我们根据众所周知的6:2:2比例划分训练集、验证集和测试集。

4.5.4 结果

我们将预训练的 RoFormer 模型应用于 CAIL2019-SCM，并使用不同的输入长度。该模型在在相同预训练数据上预训练的 BERT 和 WoBERT 模型进行比较，如表(5)所示。对于短文本截断，即 512，RoFormer 的结果与 WoBERT 相当，并且略优于 BERT 实现。然而，当将最大输入文本长度增加到 1024 时，RoFormer 的性能优于 WoBERT，绝对提升为 1.5%。

表 5：CAIL2019-SCM 任务上的实验结果。第一列中的数字表示最大截断序列长度。结果以准确率百分比表示。

模型	验证	Test
BERT-512	64.13%	67.77%
WoBERT-512	64.07%	68.10%
RoFormer-512	64.13%	68.29%
RoFormer-1024	66.07%	69.79%

4.5.5 工作局限性

尽管我们提供了理论依据以及有前景的实验证明，但我们的方法受限于以下事实：

- 尽管我们从数学上以2D子空间下的旋转形式格式化了相对位置关系，但对于为什么它比包含其他位置编码策略的基线模型收敛得更快，缺乏彻底的解释。
- 尽管我们已经证明了我们的模型在内部词对产品上具有长期衰减的有利特性，第 (3.3) 节，这与现有的位置编码机制相似，但我们的模型在长文本上的表现优于同类模型，我们还没有提出一个可靠的解释。

我们提出的RoFormer基于基于Transformer的基础设施，这需要硬件资源用于预训练目的。

5 结论

在这项工作中，我们提出了一种新的位置嵌入方法，该方法在自注意力中结合了显式的相对位置依赖性，以增强 Transformer 架构的性能。我们的理论分析表明，相对位置可以使用自注意力中的向量积自然地表示，绝对位置信息通过旋转矩阵进行编码。此外，我们通过数学方法说明了所提出的方法在应用于 Transformer 时的优势特性。最后，在英语和中文基准数据集上的实验表明，我们的方法在预训练中促进了更快的收敛。实验结果还表明，我们提出的 RoFormer 在长文本任务上可以实现更好的性能。

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